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Original Article

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Dynamic instability of breathing during sleep predicts cognition, disease, and mortality

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Abstract

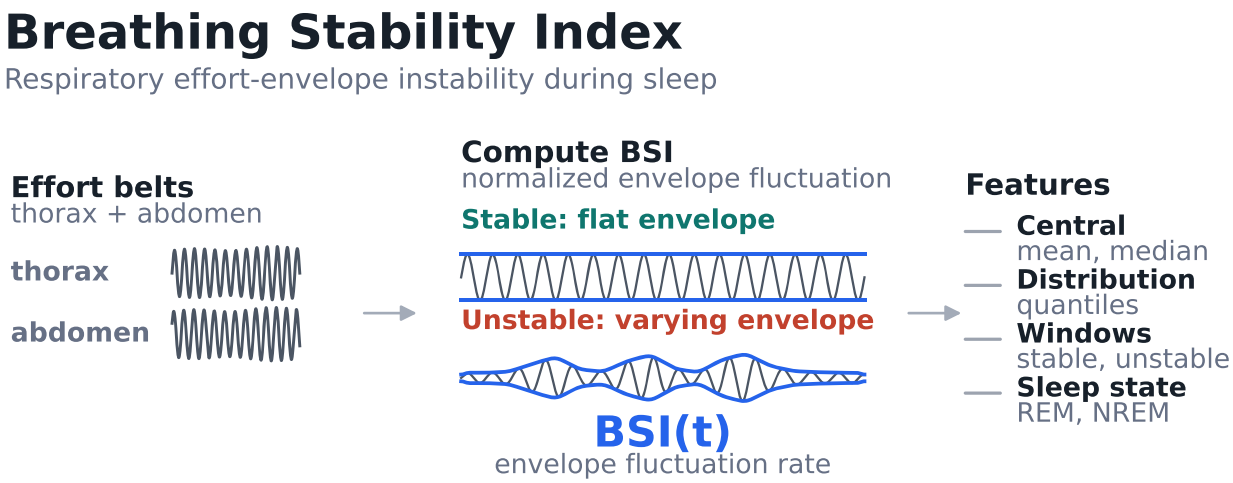
Study Objectives: To test whether an interpretable effort-derived Breathing Stability Index (BSI) captures physiology and clinical risk beyond conventional sleep-disordered breathing metrics.
Materials and Methods: We analyzed 43 611 overnight polysomnography records from two clinical cohorts and one community cohort. BSI was derived from abdominal and thoracic effort envelopes as a continuous measure of respiratory effort instability. We compared BSI with apnea–hypopnea index (AHI), hypoxic burden, arousal index, periodic breathing/self-similarity, desaturation frequency, sleep stage, and event phenotype; evaluated cognition and nine matched cross-sectional electronic-health-record (EHR) disease labels; and tested mortality in Cox models.
Results: BSI increased across AHI severity and tracked comparator metrics while remaining partly nonredundant. In mutually adjusted models, BSI remained associated with periodic breathing, AHI, arousal index, and desaturation index. Within AHI strata, instability was lowest in deeper non-rapid eye movement (NREM) sleep and higher in lighter sleep and REM sleep. Cognitive associations were cohort dependent and strongest in Osteoporotic Fractures in Men (MrOS), where less stable breathing was associated with worse fluid and total cognition after age/sex adjustment. In matched EHR disease analyses, BSI features provided modest discrimination (area under the receiver operating characteristic curve 0.509–0.642) and age/sex-adjusted associations for 8 of 9 labels. BSI was associated with all-cause mortality after adjustment for age, sex, and comparator sleep metrics in Massachusetts General Hospital (hazard ratio [HR] 1.38 per SD, 95% confidence interval [CI] 1.23–1.54) and Beth Israel Deaconess Medical Center (HR 1.10, 95% CI 1.02–1.19), but not MrOS.
Conclusions: Nocturnal breathing stability captures dynamic sleep-disordered breathing physiology that complements event counts and desaturation metrics and relates to cognition, cross-sectional disease labels, and cohort-dependent survival risk.

Key words: sleep apnea; respiration; polysomnography; respiratory instability; sleep stages; cognitive function; mortality; sleep therapy

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Graphical Abstract



Statement of Significance

Why it matters: Counting apneas and hypopneas overlooks how breathing is organized through the night. We show that a simple stability measure, quantifying steady vs erratic ventilation, adds information beyond standard indices.
What's new: Across 43 000 studies, instability patterns matched known physiology, predicted cognition, and multiple diseases better than event counts or hypoxic summaries, and tracked survival risk.
So what: Breathing stability offers a clear, interpretable lens on respiratory control with direct clinical relevance. It can help flag patients at risk even when traditional metrics are borderline and may serve as a target and surrogate outcome in interventions.

Introduction

Obstructive sleep apnea (OSA) is highly prevalent worldwide and contributes to cardiometabolic, neurocognitive, and mortality risk. Yet its standard clinical severity metric, the apnea–hypopnea index (AHI), reduces a complex disorder to an event frequency and has limited prognostic value [1–8]. This limitation has motivated alternative severity measures that separate respiratory disturbance from its consequences. Hypoxic burden integrates the depth and duration of respiratory-event-associated desaturation and predicts cardiovascular and all-cause mortality more strongly than AHI in some cohorts [9]. Ventilatory burden, an automated breath-by-breath measure of reduced airflow amplitude, also improves risk stratification and has low night-to-night variability [10, 11]. Other work has emphasized event duration, arousal burden, OSA endotypes, and ventilatory-control traits as complementary dimensions of sleep-disordered breathing [12–19] and, more broadly, sleep data including electroencephalography (EEG), electrooculography (EOG), and multimodal polysomnography (PSG) signals have supported machine-learning biomarkers for mortality, cognition, disease burden, and future disease prediction [20–22].

A related but less directly measured dimension is the *temporal organization* of breathing during sleep. Breathing can be sustained and regular for long periods, or it can fluctuate rapidly through waxing-and-waning effort, arousals, recovery breaths, event clusters, and state transitions. Rapid eye movement (REM) and non-rapid eye movement (NREM) sleep differ in respiratory drive, upper-airway muscle tone, chemosensitivity, and arousal propensity, and these state-dependent changes can alter respiratory stability even within the same AHI stratum [23–26] Existing clinical indices do not provide a simple whole-night measure of effort stability that is independent of manual event scoring.

Here we introduce the Breathing Stability Index (BSI), an interpretable, gradient-based measure derived from respiratory effort-envelope dynamics [27]. BSI yields a continuous time series of instability and summary features across sleep stages and overnight stability states. In this first large-scale polysomnography evaluation of an effort-derived breathing-stability signal, we tested whether BSI captures physiology that is not fully encompassed by AHI and hypoxic burden, and whether breathing-stability features relate to neurocognitive performance, cross-sectional disease labels, and survival risk across complementary cohorts. The goal was not to replace established metrics, but to make the temporal organization of nocturnal respiratory effort measurable for physiologic and clinical phenotyping.

Materials and methods

Study cohorts

We analyzed overnight PSG with linked phenotypic and outcome data from three study sources. The Massachusetts General Hospital clinical cohort (MGH/S0001) included adult patients

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Table 1. Cohort characteristics by BSI median sleep group

Characteristic	<0.5	0.5-1.5	>1.5
Cohort			
PSG rows, n	4664	28 436	10 511
Unique subjects, n	4382	24 183	9367
Demographics			
Age in years	48.4 (17.8)	55.9 (17.1)	62.5 (16.2)
Male sex, n (%)	2232 (47.9%)	16 523 (58.1%)	7429 (70.7%)
BMI, kg/m^2	27.90 [24.40, 32.56]	29.94 [26.10, 35.01]	29.80 [26.20, 34.58]
Sleep and respiratory features			
AHI, events/h	2.30 [0.48, 7.46]	11.24 [3.94, 22.23]	25.56 [11.59, 43.12]
Hypoxic burden, %min/h	1.00 [0.00, 5.20]	9.70 [2.40, 25.60]	28.20 [9.60, 56.80]
Arousal index, events/h	3.68 [0.43, 15.40]	16.21 [5.59, 28.56]	23.92 [11.78, 39.14]
Desaturation index, events/h	5.18 [1.93, 11.54]	15.61 [7.30, 28.16]	31.51 [16.88, 52.02]
Primary exposures			
BSI median during sleep	0.37 [0.26, 0.44]	1.01 [0.77, 1.25]	1.73 [1.60, 1.92]
BSI quantile 75 during sleep	0.66 [0.52, 0.78]	1.49 [1.21, 1.74]	2.23 [2.07, 2.48]
BSI quantile 90 during sleep	1.03 [0.83, 1.26]	1.97 [1.65, 2.24]	2.72 [2.51, 3.04]
BSI instability burden >1.5 (% of total sleep)	1 [0, 5]	58 [26, 86]	100 [97, 100]
Periodic breathing (% of total sleep)	0.3 [0.0, 1.2]	1.9 [0.5, 4.9]	5.4 [1.7, 12.4]
Phenotype			
OSA phenotype, n (%)	883 (18.9%)	8927 (31.4%)	3794 (36.1%)
CSA phenotype, n (%)	1627 (34.9%)	8546 (30.1%)	2892 (27.5%)
Mixed phenotype, n (%)	708 (15.2%)	6916 (24.3%)	2784 (26.5%)

Values are shown as n, n (%), n/N (%), mean (SD), or median [IQR], as appropriate. Abbreviations: AHI, apnea-hypopnea. Cardiometabolic comorbidities are shown only where binary covariate data were available for the primary mortality-model adjustment set. Values are present/assessed n/N (%). Depression is from a separate cross-sectional diagnosis flag and therefore has a separate assessed denominator. OSA/CSA phenotype: OSA if OAI/(OAI + CAI) > 0.75, CSA if <0.25, mixed if 0.25–0.75.

referred for in-laboratory PSG from 2010 to 2023. Subsets of these patients had linked neuropsychological testing, electronic-health-record (EHR) diagnosis fields and vital-status follow-up, and/or EHR-derived survival follow-up. The Beth Israel Deaconess Medical Center clinical cohort (BIDMC/I0002) included adult patients referred for in-laboratory PSG at the BIDMC sleep laboratory with linked survival follow-up [28–31]. The Osteoporotic Fractures in Men Study (MrOS) included community-dwelling men aged 65 years or older who underwent unattended home PSG with subsequent survival follow-up [32, 33].

Analyses used endpoint-specific analysis sets because available outcomes differed by cohort. Table 1 summarizes the physiology-ready analysis population by BSI category, and Table S1 reports cohort-specific availability for cognition, disease labels, mortality follow-up, and key covariates. Descriptive summaries are PSG-level unless otherwise specified. Mortality analyses used one PSG per participant from MrOS, BIDMC, and MGH to test whether multi-feature breathing-stability profiles were associated with survival risk. Institutional review boards at BIDMC (#2024P000804) and MGH (#2013P001024) approved the study with a waiver of informed consent for secondary analyses; all other cohorts were accessed under executed data-use agreements and local approvals.

Cognition and disease outcomes

Cognitive analyses used MGH and MrOS subsets. In MGH-Cog, neuropsychological testing was performed within 30 days of PSG. In MrOS, cognitive measures were obtained from the MrOS Sleep Study cycle corresponding with the PSG and were typically obtained within 2 years of the sleep recording. All cognition analyses were treated as cross-sectional.

Disease and comorbidity phenotypes were derived from EHR-based phenotyping in the MGH subset, as previously described [28]. Briefly, disease phenotypes used validated EHR rules combining International Classification of Diseases (ICD) diagnosis codes, problem-list entries, relevant medication records, cognitive screening scores for dementia/mild cognitive impairment (MCI) when available, and targeted clinician-vetted text searches. Disease diagnoses were defined as present at or before the time of PSG; therefore, disease analyses were cross-sectional with respect to the sleep study rather than incident disease analyses. We evaluated nine disease categories: dementia, mild cognitive impairment, symptomatic disease, atrial fibrillation, myocardial infarction, type II diabetes, hypertension, bipolar disorder, and depression. Disease-specific case and control counts are reported in Table S2.

For each disease, cases were matched 1:1 to controls using exact sex matching and nearest-age matching without replacement. Controls were selected from participants without any of the evaluated disease diagnoses, with matching performed separately for each disease. Age balance after matching was assessed using standardized mean differences and two-sample Kolmogorov–Smirnov tests.

Mortality outcomes

Mortality analyses were restricted to participants with longitudinal vital status and follow-up time. Mortality status was obtained from cohort-specific sources, including hospital EHR mortality records and National Death Index linkage when available. Participants were followed from PSG until death or the last available follow-up/censoring date. Cohort-specific follow-up availability, mortality data source, follow-up time, and event counts are summarized in Table S3. Transparent median-BSI mortality models used one diagnostic or non-treatment PSG per participant to avoid mixing diagnostic and treatment physiology; complementary score-based analyses used one PSG per participant to test whether multifeature breathing-stability profiles related to survival risk.

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Conventional respiratory severity metrics

AHI was defined as scored apneas plus hypopneas per hour of sleep and categorized as normal (<5), mild (5 to <15), moderate (15 to <30), or severe (> = 30) events/h. Hypoxic burden was calculated as the sleep-time-normalized area under respiratory-event-associated oxygen desaturation curves. Desaturation index (3% oxygen desaturations) was retained as a frequency-based desaturation comparator. Arousal index was defined as scored EEG arousals per hour of sleep. Periodic breathing was quantified using a self-similarity algorithm that identifies recurring stereotyped respiratory flow-reduction patterns [13, 27]; for interpretability, we report the percentage of sleep tagged as self-similar as periodic breathing (%), while retaining self-similarity as the technical algorithm name.

Signal preprocessing

Routine PSG followed American Academy of Sleep Medicine standards and included thoracic and abdominal respiratory inductance plethysmography belts, oronasal flow, and oximetry [34]; sleep stages were manually scored in 30-s epochs by registered technologists. For BSI, available abdominal and thoracic effort channels were selected, detrended, low-pass filtered at 10 Hz, and resampled to 10 Hz. Each belt was then robust-normalized separately using the median and interquartile range after clipping extreme amplitudes at the 2nd and 98th percentiles. When both belts were available, the primary effort signal was defined as the average of the robust-normalized abdominal and thoracic signals; when only one usable belt was available, that belt was used.

To capture the slowly varying amplitude modulations of breathing, we extracted upper (up) and lower (lo) envelopes from the effort trace in the following steps:

1) Peak and trough detection. Local maxima and minima were identified using a peak-finding algorithm with a minimum inter-extrema interval of 5 s (≈ 3–8 breaths, depending on rate). A moving-average baseline (10 s window) was used to exclude spurious extrema: candidate maxima were required to exceed the baseline by at least 10%, and minima to fall below 90% of baseline.
2) Spline interpolation. The sets of accepted peaks and troughs were interpolated across the full signal length using cubic splines, yielding smooth estimates of the upper and lower amplitude envelopes.
3) Outlier removal. Extreme peaks or troughs inconsistent with neighboring values were detected using a median absolute deviation criterion (threshold factor = 5). Outliers were replaced by the nearest valid neighbor, and the corresponding spline interpolation recomputed. This step prevents isolated artifacts (e.g. belt slippage or motion) from biasing the envelope.
4) Artifact correction. After interpolation, envelopes were constrained to maintain the physiological ordering (up ≥ lo). Instances of envelope crossing or implausible values relative to the raw trace were corrected by local replacement with the running mean of preceding samples.
5) Smoothing. Final envelopes were lightly smoothed using cubic smoothing splines to emphasize the slow modulations of breathing amplitude while preserving cycle-level dynamics.

This procedure robustly captures the waxing-and-waning morphology of respiratory effort, emphasizing physiologically relevant amplitude variability while minimizing contamination by artifacts. The approach builds on prior work by Oppersma *et al.* [27, 35, 36] but differs in: (i) it relies on explicit peak/trough detection with physiologically informed exclusion rules rather than analytic-signal envelopes, (ii) it applies robust outlier detection and correction, and (iii) it enforces consistent upper–lower envelope ordering throughout the signal.

Breathing stability index

BSI is a dimensionless measure of dynamic respiratory envelope variability. It is computed as the sum of the mean absolute gradients of the upper and lower respiratory effort envelopes, normalized by their local mean amplitude. This yields a volatility-like signal reflecting how rapidly and strongly the envelope modulates relative to its baseline. A value of 0 corresponds to no envelope variation (i.e. flat or constant respiratory effort), while a value of 1 indicates that the envelope fluctuates by ∼100% of its local amplitude baseline—analogous to a normalized coefficient of variation per unit time.

We define the BSI in the following steps:

1) Envelope extraction.
   Let $up(t)$ and $lo(t)$ denote the upper and lower amplitude envelopes of the respiratory effort signal.
2) Envelope fluctuation rate (numerator).

Within a symmetric window of duration Δ centered at time t, compute the mean absolute temporal gradient:

$$G(t) = \frac{1}{\Delta}\int_{t-\Delta/2}^{t+\Delta/2} \left|\frac{d}{dt'} up\,(t')\right| + \left|\frac{d}{dt'} lo\,(t')\right| \; dt'$$

3) Local amplitude normalization (denominator).

$$A(t) = \frac{1}{\Delta}\int_{t-\Delta/2}^{t+\Delta/2} \frac{1}{2}\left(up\,(t') + lo\,(t')\right) \; dt'$$

4) Final BSI definition.

$$\mathbf{BSI}(t) = \frac{G(t)}{A(t)}$$

The methodology for computing the BSI is shown in Figure 1 (panels A–D). We prespecified and set $\Delta = 120$ s and computed the BSI time-series using sliding windows with 90% overlap (i.e. a 12-s step size). The 90% overlap was used as dense sliding-window sampling to provide smoother coverage across adjacent epochs and reduce dependence on arbitrary window boundaries.

The resulting BSI captures moment-to-moment breathing instability, with higher values indicating faster or larger envelope modulations. Example BSI time-series overlaid on overnight respiratory effort is shown in Figures 2 and 3, illustrating transitions between stable and unstable breathing regimes.

Three families of nightly features were derived: central tendency (median BSI), distributional features (area under the curve, inter-decile range, and quantiles), and state counts. State counts were the numbers of non-overlapping 5-min windows that were stable (all BSI <0.50) or unstable (all BSI >1.50). Feature computation was repeated separately for all sleep, NREM, REM, and individual N1/N2/N3 stages to capture state-dependent physiology.

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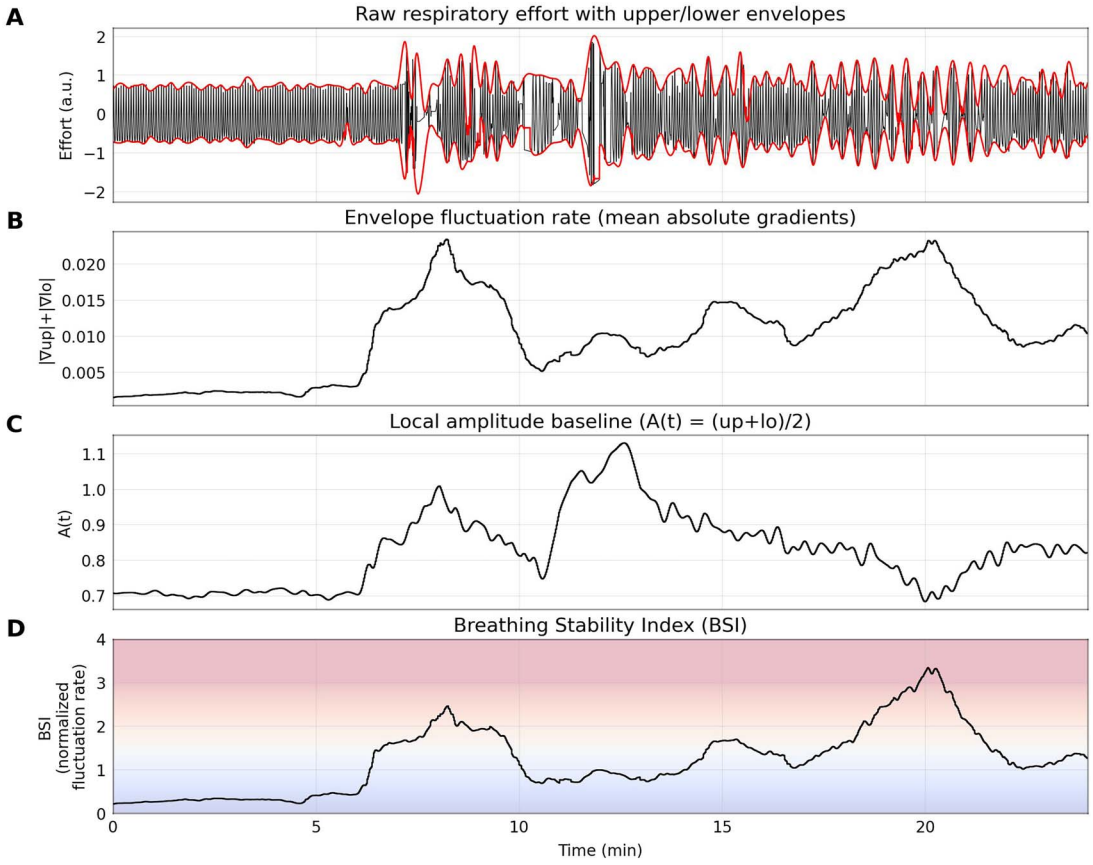


Figure 1. Construction of the BSI. The BSI quantifies how strongly and how rapidly respiratory effort amplitude fluctuates relative to its local baseline. (A) Example respiratory effort tracing with automatically derived upper and lower envelopes. (B) Envelope fluctuation rate, computed as the mean absolute gradients of the envelopes, increases during waxing/waning breathing patterns. (C) Local amplitude baseline, defined as the mean of the upper and lower envelopes, provides normalization for breathing strength. (D) The BSI is the ratio of envelope fluctuation rate to baseline amplitude, yielding a normalized fluctuation rate (dimensionless). Low values indicate stable, regular breathing; high values mark unstable regimes with pronounced amplitude modulation. See Methods for details.

Physiologic validation

To evaluate physiologic interpretation, we compared mean sleep BSI with established and complementary PSG-derived respiratory measures, including periodic breathing/self-similarity, AHI, hypoxic burden, arousal index, and desaturation index/ODI3. For each comparator, we visualized the relationship with mean sleep BSI using hexbin density plots and smoothed nonlinear trends. We report Spearman correlations and standardized BSI coefficients from linear models adjusted for age, sex, BMI, and cohort. In secondary models, we additionally adjusted each comparator association for the other respiratory comparators to assess overlap across physiologic domains.

To examine state-dependent variation, we computed BSI separately within N1, N2, N3, REM, and combined NREM sleep. We first tested whether BSI increased across AHI severity categories within each sleep stage using linear mixed-effects models with BSI as the outcome, AHI category as the fixed effect, and a random intercept for participant. We then tested whether BSI differed by sleep stage within each AHI severity category using mixed-effects models with sleep stage as the fixed effect and a random intercept for PSG. We report the omnibus *p*-value and marginal R-squared (R2m), which quantifies variance explained by the fixed effect. *p*-values for the nine planned omnibus tests were adjusted using Holm correction.

We classified respiratory-event phenotype using the obstructive apnea index (OAI) and central apnea index (CAI). The obstructive fraction was defined as OAI/(OAI + CAI), with obstructive sleep apnea (OSA) classified as >0.75, central sleep apnea (CSA) as <0.25, and mixed otherwise. We compared median sleep BSI across phenotypes overall and within AHI categories, using CSA as the reference in adjusted linear models.

Cognitive and disease prediction

For each cognitive outcome and cross-sectional disease label, we derived a target-specific breathing-stability score using only the prespecified BSI feature block: sleep, NREM, and REM quantiles plus 5-min stable/unstable window counts. Scores were generated out of fold using ridge regression for cognition and ridge-penalized logistic regression for disease, with subject-aware folds for cognition and matched-pair-aware folds for disease. Prediction performance was summarized as out-of-fold Pearson correlation for cognition and area under the receiver operating characteristic curve (AUROC) for disease, with age/sex demographic benchmarks computed using the same fold structure.

For cognition, standardized age- and sex-adjusted linear models were fit separately in MrOS and MGH-Cog. For disease, age- and sex-adjusted logistic models were fit in the matched cross-sectional case–control sets, with standard errors clustered by matched pair. Continuous exposures were standardized, and cognition effects were oriented so that positive beta values indicate worse cognition per 1 SD less favorable physiology; disease effects are reported as odds ratios per 1 SD higher target-specific breathing-stability risk score.

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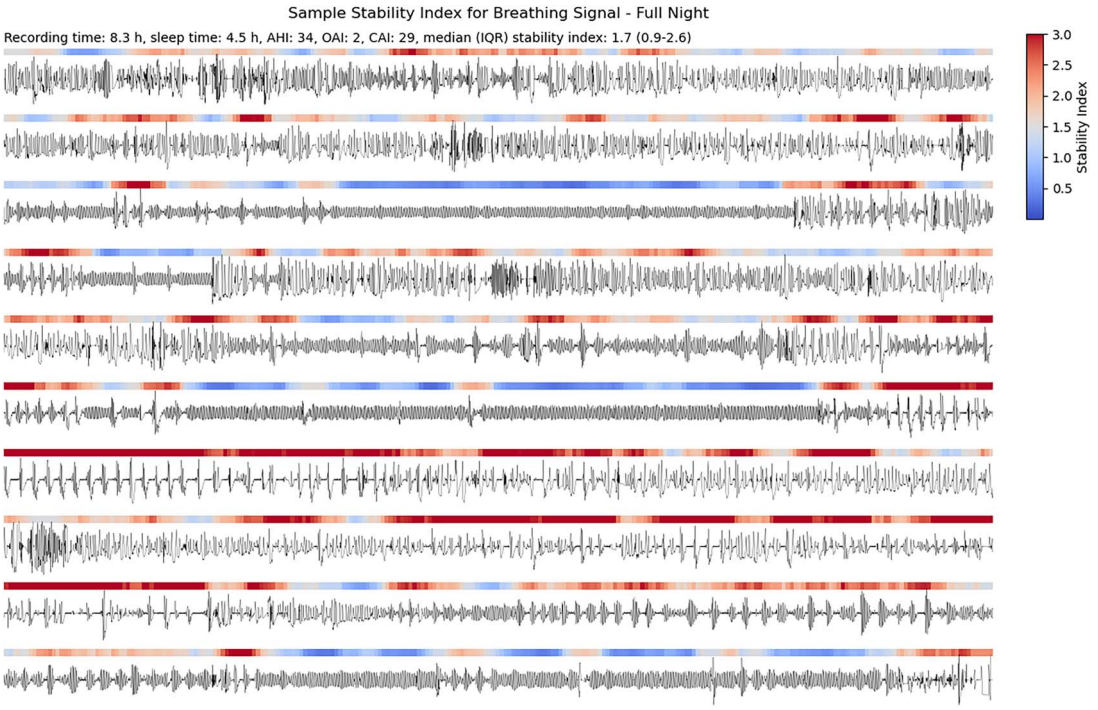


Figure 2. Example night with pronounced instability. Overnight respiratory effort (summed thoraco-abdominal belts) with envelopes and the BSI overlaid. BSI is computed from the mean absolute gradients of the upper/lower effort envelopes normalized by local amplitude ($\Delta = 120$ s windows, 90% overlap); larger values denote faster/greater envelope modulation. In this participant, clusters of apneas/hypopneas and irregular ventilation coincide with sustained elevations in BSI (frequent excursions >1.5), whereas brief quiescent periods show near-baseline BSI (<0.5).

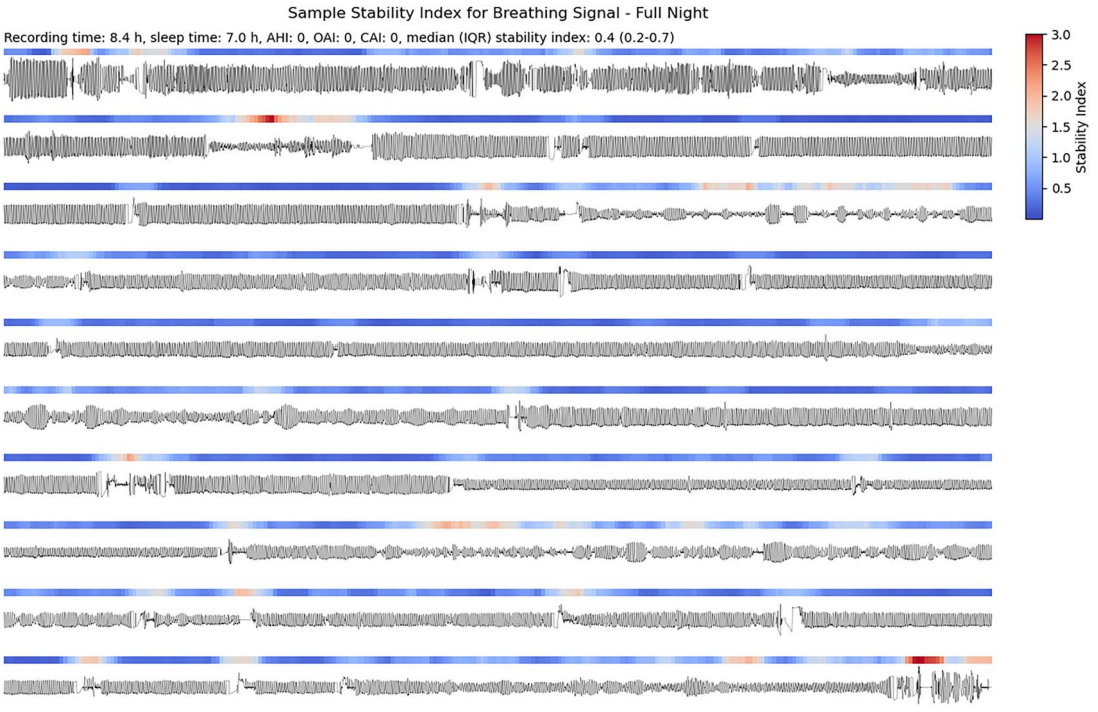


Figure 3. Example night with predominantly stable breathing. Contrasting example to Fig. 2. Most of the night shows organized ventilation with low, steady BSI values (frequent 5-min windows with BSI < 0.5), punctuated by brief elevations during transitional periods. The same computation ($\Delta = 120$ s, 90% overlap) and thresholds (0.5, 1.5) are applied here, highlighting that BSI separates nights dominated by stability from those dominated by instability independent of manual event scoring.

Survival analysis

For mortality, breathing stability was modeled as a multifeature exposure. We generated out-of-fold BSI mortality scores from prespecified sleep, NREM, and REM quantiles and stable/unstable 5-min window features. Within each training fold, a ridge-penalized Cox model was fit using BSI features and the covariates specified for that model; the BSI-feature component was then applied to held-out participants to generate an

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out-of-fold breathing-stability score. Cohort-specific Cox models tested these scores with hazard ratios reported per 1-SD increase. Models were fit sequentially as unadjusted, adjusted for age and sex, additionally adjusted for AHI, and fully adjusted for age, sex, AHI, hypoxic burden, arousal index, and periodic breathing/self-similarity. Comparator fully-adjusted models also included the cohort-specific out-of-fold BSI mortality score. BMI was evaluated as an additional sensitivity covariate when available. Sex was omitted when constant within a cohort.

Sensitivity analyses

Supplementary analyses evaluated several processing and design questions. We compared primary diagnostic/untreated PSGs with positive airway pressure (PAP)/titration, split-night, and repeat diagnostic PSGs when multiple studies were available. We evaluated REM and NREM BSI separately, the REM-NREM BSI difference, and a REM-predominant instability flag for mortality and cross-sectional total cognition. We assessed 1-, 2-, and 5-min BSI windows and performed a targeted overlap experiment at 50%, 75%, 90%, 95%, and 99% overlap in representative files. We also examined posture availability and supine burden in cohorts with body-position fields. Finally, we identified representative paradoxical-breathing segments, recomputed local BSI using abdomen only, thorax only, and the current combined-belt implementation, and visualized these examples in the Supplement (Figure S1).

Results

Study population

Complete respiratory envelopes and linked phenotypes were available for 43 611 PSG records, including 24 866 MGH/S0001 in-laboratory PSGs, 14 788 BIDMC in-laboratory PSGs, 3768 MrOS unattended home studies, and 189 MGH-Cog studies. Cohort characteristics are summarized in Table 1 and Table S1. MGH participants were middle-aged (53.2 +/– 16.7 years; 42.9% female) with predominantly mild sleep-disordered breathing (mean AHI, 12.5 +/– 14.8 events/h). BIDMC participants were slightly older (57.3 +/– 16.7 years; 44.9% female) and had higher AHI (23.1 +/– 20.1 events/h). MrOS participants were older (77.5 +/– 5.6 years), all male, and had more severe sleep apnea (31.4 +/– 18.8 events/h).

The BSI mortality-score analyses were run after one-row-per-participant and feature/covariate eligibility filters, yielding 2453 MrOS participants with 1912 deaths over a median follow-up of 12.2 years (IQR, 7.7–17.7), 6475 BIDMC participants with 709 deaths over a median follow-up of 3.0 years (inter-quartile range [IQR], 1.3–4.7), and 8266 MGH participants with 353 deaths over a median follow-up of 5.4 years (IQR, 2.7–8.3).

The disease prediction analyses used disease-specific 1:1 age- and sex-matched case–control samples from the EHR-phenotyped MGH subset. Final matched sample sizes were 774 for dementia, 1170 for MCI, 3846 for symptomatic disease, 1250 for atrial fibrillation, 860 for myocardial infarction, 728 for type II diabetes, 5338 for hypertension, 466 for bipolar disorder, and 2134 for depression. Matching produced near-identical age distributions between cases and controls for all diseases (Table S2), reducing age-sex confounding in the disease classification analyses.

Distribution and face validity of BSI

BSI produced a continuous, dimensionless time series whose excursions visually coincided with clusters of apnea, hypopnea, recovery breathing, and other unstable effort patterns (Figures 2 and 3). Across all nights, the median whole-night BSI was 1.14 (IQR, 0.78-1.60). Participants spent an average of 2.2 +/– 2.8 h in highly stable breathing, defined as continuous 5-min windows with BSI <0.50, and 3.4 +/– 3.2 h in pronounced instability, defined as BSI >1.50. Thus, BSI translated routinely recorded effort belts into an interpretable overnight map of stable and unstable respiratory-control states.

Association with AHI severity and sleep stages

Mean sleep BSI showed a coherent physiologic gradient with periodic breathing, AHI, hypoxic burden, and arousal index (Spearman rho = 0.438, 0.472, 0.450, and 0.327; demographic-adjusted beta = 0.504, 0.484, 0.163, and 0.342, respectively; Figure 4). In mutually adjusted models including the original respiratory comparators, associations persisted for periodic breathing, AHI, and arousal index (beta = 0.234, 0.217, and 0.210; all $p < 1e-200$), and were markedly attenuated but still significant for hypoxic burden (beta = 0.017, $p = .023$). In a parallel sensitivity model, BSI also remained associated with ODI3/desaturation index after adjustment for periodic breathing, AHI, hypoxic burden, arousal index, age, sex, BMI, and cohort (beta = 0.081, $p = 3.2e-28$). Higher BSI also tracked fewer consecutive stable 5-min windows and more consecutive unstable 5-min windows (rho = –0.841 and 0.899), supporting BSI as a non-redundant measure of respiratory effort organization rather than a simple recapitulation of event counts or oxygen desaturation.

Stage-specific analyses showed that BSI increased across standard AHI severity categories within each sleep stage (Figure 5A). In mixed-effects models accounting for repeated studies within participants, AHI category explained a substantial fraction of stage-specific BSI variation in NREM sleep (R2m = 0.21), N1 (R2m = 0.20), and N2 (R2m = 0.18), with smaller but significant effects in N3 (R2m = 0.06) and REM (R2m = 0.10). All Holm-adjusted omnibus p-values were < 1e–6.

BSI also varied systematically by sleep stage within each AHI severity category (Figure 5B). Across AHI strata, N3 had the lowest instability and N1 the highest instability, consistent with state-dependent respiratory control. Sleep stage explained 10% of BSI variation in the normal AHI group (R2m = 0.10), 16% in mild AHI (R2m = 0.16), 17% in moderate AHI (R2m = 0.17), and 15% in severe AHI (R2m = 0.15), with all Holm-adjusted omnibus p-values <1e–6. Together, these findings show that BSI tracks expected gradients in AHI severity while also capturing sleep-state physiology that is not conveyed by a whole-night event frequency alone.

Event-phenotype analyses supported interpreting BSI as a broad effort-instability measure rather than an OSA-specific metric. BSI differed across OSA, CSA, and mixed phenotypes before full respiratory-burden adjustment, and within each phenotype median BSI increased with AHI severity (Supplement). After adjustment for AHI and hypoxic burden, phenotype contrasts attenuated, indicating that BSI reflects shared instability expressed across obstructive, central, and mixed event patterns.

Supplementary REM/NREM outcome analyses showed similar interpretation. REM and NREM BSI could be summarized separately, but the REM-NREM BSI difference and REM-predominant instability flag were not independently associated with mortality or cross-sectional total cognition after covariate adjustment (Table S7).

Breathing instability and neurobehavioral/disease state outcomes

Breathing-stability features showed endpoint- and cohort-dependent out-of-fold prediction of cognition, with the strongest and most consistent performance in MrOS. In MrOS, BSI-feature

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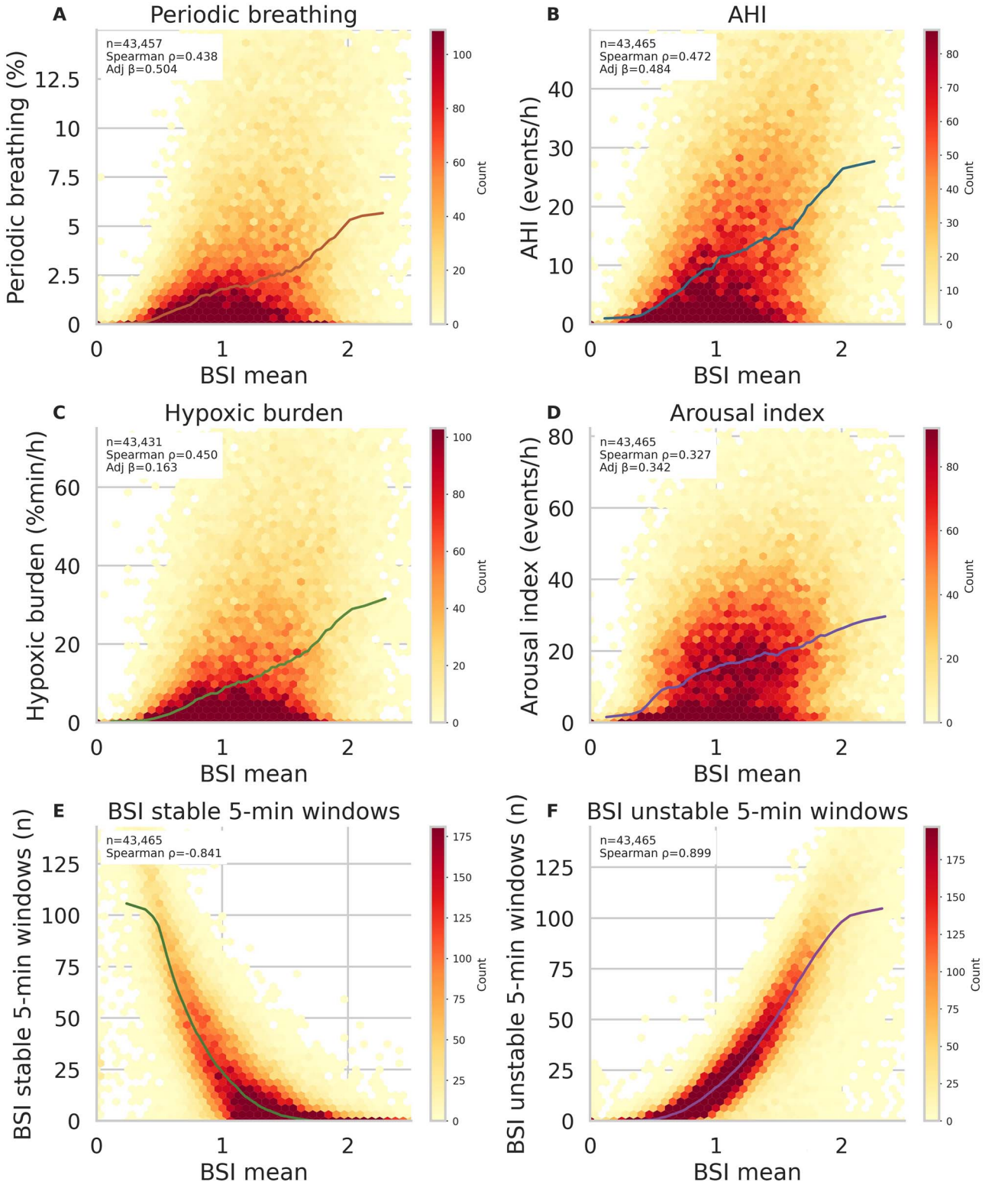


Figure 4. Mean sleep BSI tracks respiratory instability across complementary physiologic domains. Higher mean sleep BSI was associated with greater periodic breathing, AHI, hypoxic burden, and arousal index, with positive demographic-adjusted associations for each comparator. As an internal validation of the metric, higher BSI also corresponded to fewer consecutive stable 5-min windows and more consecutive unstable 5-min windows. Hexbin colors indicate recording density; curves show smoothed nonlinear trends. Panel annotations show sample size, Spearman ρ, and standardized BSI β adjusted for age, sex, BMI, and cohort.

scores predicted fluid, crystallized, and total cognition with Pearson r values of 0.109, 0.049, and 0.129, respectively, although adding BSI features to age and sex changed prediction only minimally. In MGH-Cog, BSI features predicted fluid cognition ($r = 0.188$), whereas performance for crystallized and total cognition was weak (Figure 6, Tables S4 and S5).

In age- and sex-adjusted association models, less favorable breathing stability was associated with worse fluid cognition

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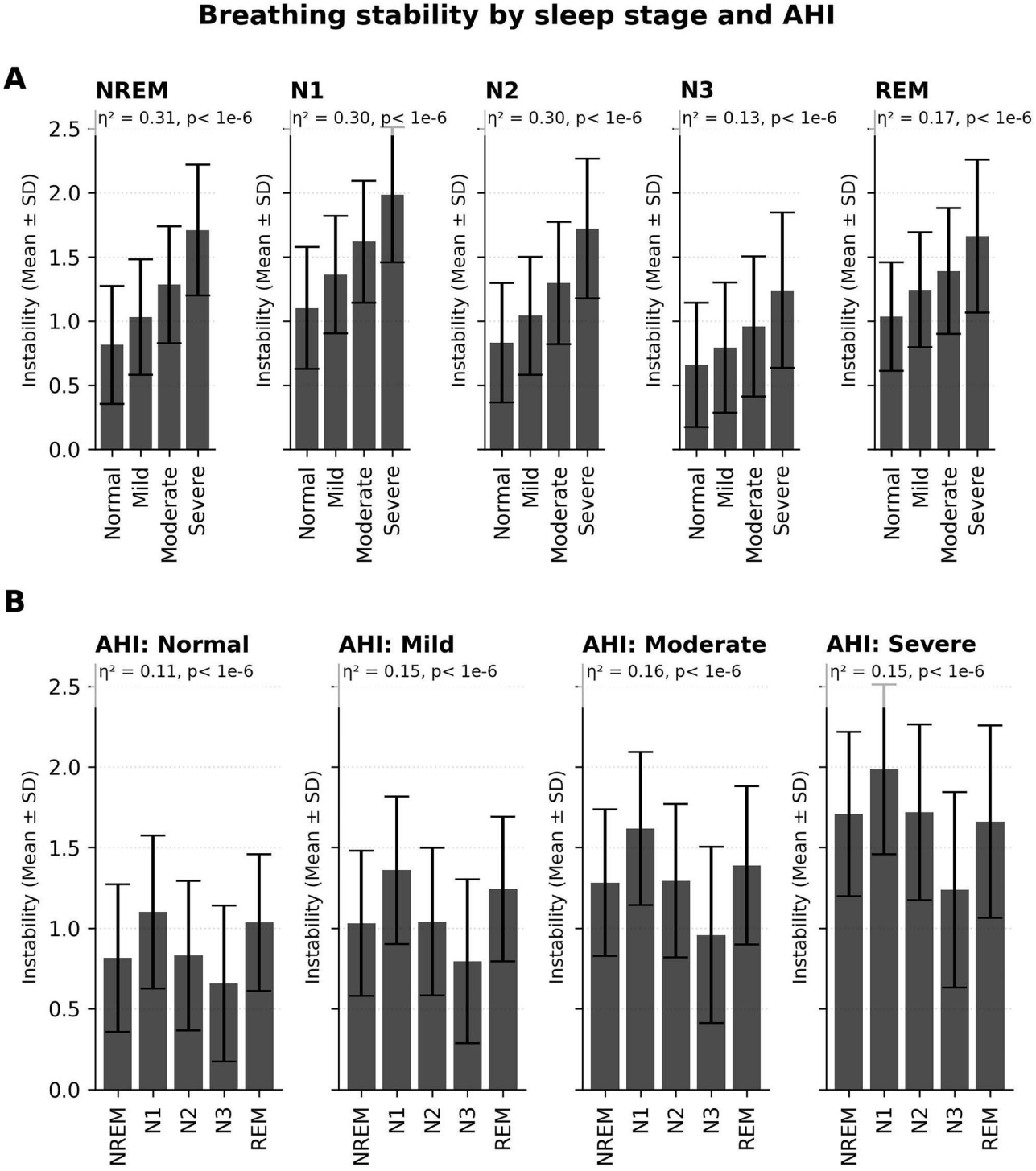


Figure 5. Association of BSI with AHI severity and sleep stage. (A) Stage-specific BSI by AHI severity category. Bars show mean ± SD BSI within each sleep stage across standard AHI categories: Normal (<5 events/h), mild (5 to <15), moderate (15 to <30), and severe (≥30). Mixed-effects models tested whether BSI differed across AHI categories within each stage, with random intercepts for participant. R^2m indicates the proportion of BSI variance explained by AHI category alone. (B) BSI by sleep stage within each AHI severity category. Bars show mean ± SD BSI across NREM, N1, N2, N3, and REM sleep. Mixed-effects models tested whether BSI differed across sleep stages within each AHI category, with random intercepts for PSG. R^2m indicates the proportion of BSI variance explained by sleep stage alone. All shown omnibus *p*-values were Holm-adjusted across the nine planned panel-level tests.

in MrOS (beta = 0.048, 95% CI 0.017–0.079) and worse total cognition (beta = 0.075, 95% CI 0.043–0.108), but not clearly with crystallized cognition. In MGH-Cog, the breathing-stability score was not significantly associated with cognition, whereas arousal index showed consistent associations with worse cognition across all three cognitive domains.

For cross-sectional disease labels, the age/sex benchmark performed at chance as expected from the matched design, whereas BSI features showed discrimination across most disease endpoints despite age/sex matching. BSI-feature AUROC ranged from 0.509 for MCI to 0.642 for atrial fibrillation. In age- and sex-adjusted matched logistic models, higher target-specific BSI

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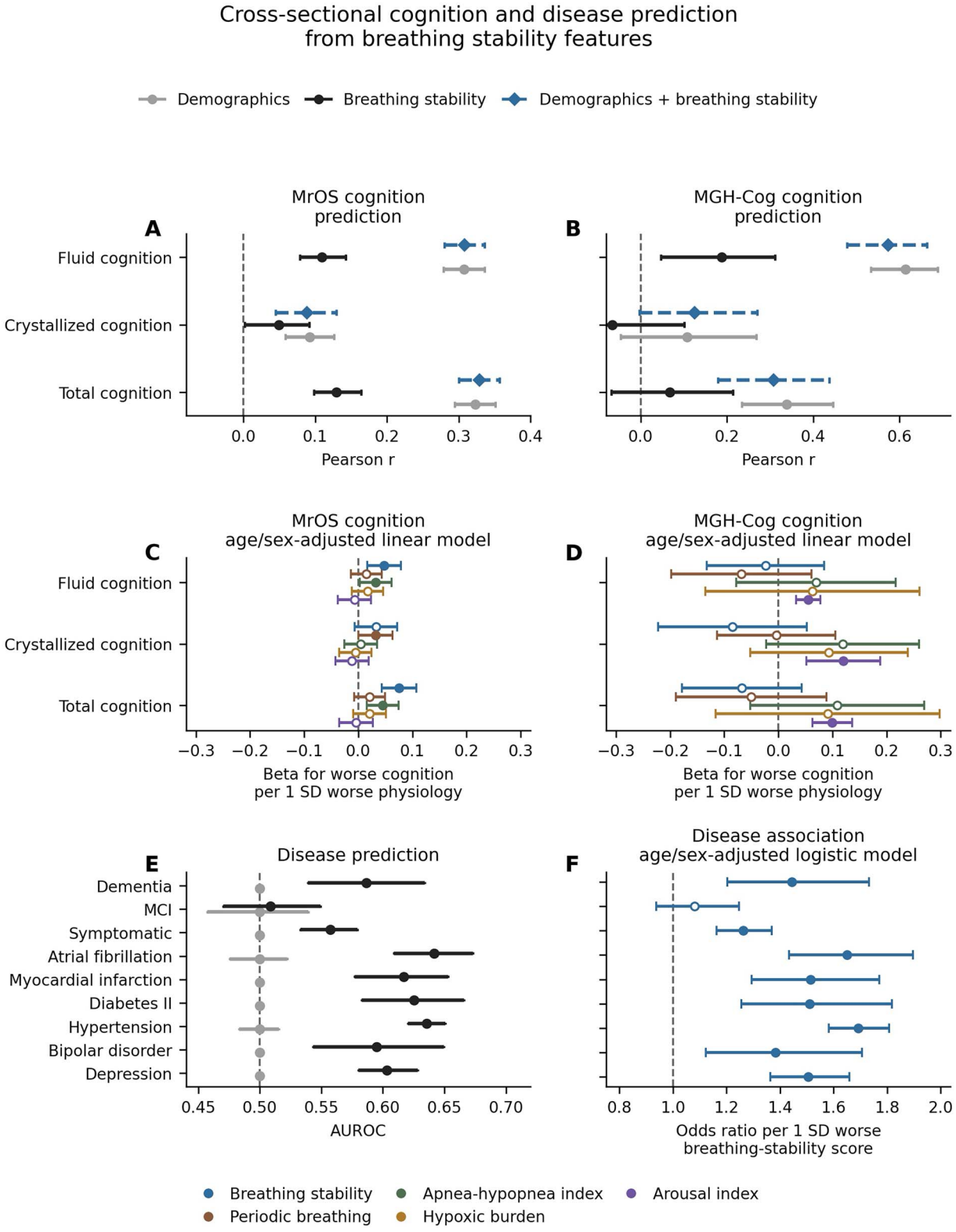


Figure 6. Cross-sectional cognition and disease prediction from breathing-stability features. Panels A and B show out-of-fold cognition prediction in MrOS and MGH-Cog using demographics, breathing-stability features, or both. Panels C and D show age- and sex-adjusted standardized associations with cognition, oriented so positive beta values indicate worse cognition per 1 SD less favorable physiology. Panel E shows disease-label discrimination by demographics and breathing-stability features in matched case-control sets. Panel F shows age- and sex-adjusted odds ratios for cross-sectional disease labels per 1 SD higher target-specific breathing-stability risk score.

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risk scores were associated with higher odds of 8 of 9 disease labels, including atrial fibrillation, hypertension, myocardial infarction, diabetes, depression, dementia, bipolar disorder, and symptomatic status; MCI was the only endpoint without a clear association.

All-cause mortality

Multi-feature breathing-stability scores showed cohort-dependent associations with all-cause mortality (Figure 7, Table S6). In MrOS, the BSI feature score was associated with mortality before adjustment (HR 1.15 per SD, 95% CI 1.10–1.20, $p = 3.1e{-}11$), but the association was attenuated after age adjustment and respiratory-comparator adjustment (fully adjusted HR 1.02, 95% CI 0.97–1.06, $p = 0.497$). In BIDMC, the out-of-fold BSI feature score was associated with higher mortality risk in unadjusted analysis (HR 1.25, 95% CI 1.16–1.35, $p = 3.8e{-}09$), after age and sex adjustment (HR 1.15, 95% CI 1.07–1.24, $p = 2.5e{-}04$), after additional AHI adjustment (HR 1.12, 95% CI 1.04–1.21, $p = 0.004$), and after adjustment for age, sex, AHI, hypoxic burden, arousal index, and periodic breathing/self-similarity (HR 1.10, 95% CI 1.02–1.19, $p = 0.012$). In MGH, the association was strongest and remained significant after full adjustment (unadjusted HR 1.83, 95% CI 1.64–2.05, $p = 1.3e{-}26$; fully adjusted HR 1.38, 95% CI 1.23–1.54, $p = 4.5e{-}08$). Thus, breathing-stability features captured survival-relevant information in both clinical cohorts, with weaker independent signal in the older MrOS community cohort.

Supplementary study-type and processing sensitivities

In paired study-type sensitivity analyses, BSI differed minimally between diagnostic and PAP/titration PSGs (median paired difference 0.014, $p = 0.125$) or split-night PSGs (median paired difference 0.0155, $p = 0.120$). Repeat diagnostic PSGs showed moderate within-subject stability with expected night-to-night variation (Spearman rho = 0.437; Table S8).

Window and overlap analyses supported the stability of subject-level median BSI summaries across the tested settings. At 90% overlap, 1-, 2-, and 5-min window summaries were highly correlated (1 vs 2 min rho = 0.995; 2 vs 5 min rho = 0.989; 1 vs 5 min rho = 0.995). In a targeted 10-file overlap experiment, median BSI at 50%, 75%, 95%, and 99% overlap was highly correlated with the 90% reference (rho range 0.988–1.000), with expected increases in computation time at denser overlap (Tables S9 and S10).

Discussion

We developed BSI to quantify temporal instability in thoraco-abdominal respiratory effort across the night. Rather than counting discrete scored events or integrating oxygen desaturation, BSI summarizes how rapidly and strongly respiratory effort, which reflects neural respiratory drive, fluctuates relative to its local baseline. Across clinical and community PSG cohorts, BSI showed strong physiologic validity: it increased with AHI severity, periodic breathing, hypoxic burden, desaturation frequency, and arousal index, while importantly remaining only partly overlapping with these measures. BSI also varied systematically by sleep stage, with lower instability in deeper NREM sleep and higher instability in lighter sleep and REM sleep. These findings support BSI as a complementary description of sleep-disordered breathing, not as a replacement for established event- or consequence-based metrics [2, 9, 10, 34, 37–42].

A central point is that BSI measures stable versus unstable effort-envelope dynamics, rather than normal versus reduced effort. Low BSI can occur during sustained, regular breathing even when effort is reduced or flow limitation is present; high BSI reflects rapid or large changes in effort amplitude, irregular effort, waxing-and-waning patterns, or transient cessations. This distinction helps explain why BSI is related to AHI and hypoxic burden but is not simply a transformed version of either. Event counts, oxygen desaturation burden, arousal burden, periodic breathing, and effort-envelope instability each describe a different aspect of the same physiologic system.

The sleep-stage findings provide important support for the metric. Instability was lowest in deeper NREM sleep and higher in lighter sleep and REM sleep, consistent with known state-dependent changes in respiratory control, upper-airway muscle tone, chemosensitivity, and arousal propensity [23–25]. Prior cardiopulmonary coupling work also supports the idea that stable breathing is linked to more organized sleep physiology [26]. In this context, BSI appears sensitive to both trait disease burden and state-dependent control. The supplementary REM/NREM analyses were consistent with this interpretation: REM and NREM BSI could be summarized separately, but the REM-NREM difference and REM-predominant instability flag did not add clear independent outcome information in the available data.

The importance of breathing stability as an integrated measure of health of widespread interconnected networks may be explained by widespread synchronizing effects of the signal [43]. The respiratory rhythm, via a centrifugal corollary discharge, acts as a functional oscillatory scaffold and provides a unifying global temporal coordination of neuronal firing and network dynamics across cortical and subcortical networks during offline states [44]. Respiration is thus a rhythmic timing signal to the brain and can be recorded at multiple sites including the hippocampus. This hippocampal respiratory rhythm (HRR) is most prominent in the dentate gyrus, especially when respiration is slower than theta frequency. Discharges of neurons in CA1 and dentate gyrus are modulated by both HRR and theta [45]. During offline states, the respiratory modulation underlies the coupling of hippocampal sharp-wave ripples and cortical DOWN/UP state transitions, which mediates systems memory consolidation [44].

Stable sleep and stable breathing may reinforce one another [43, 46]. Stable breathing periods have long been recognized during sleep, particularly in N3 and much of N2 in health, and may occur across the full range of sleep-apnea severity. These periods likely reflect coordinated electrocortical, respiratory, and autonomic network states rather than only local neuromuscular mechanics [47]. Stable breathing aligns with sleep features often considered restorative, including vagal predominance, blood-pressure dipping, fewer phasic EEG events, high-frequency cardiopulmonary coupling, and greater slow-wave activity [47]. Besides corollary effects of respiratory generative discharges, waves of sensory activity evoked by nasal airflow spread through brain circuits, synchronizing neuronal populations to the breathing cycle and modulating faster oscillations, cell assembly formation and cross-area communication, thereby providing a mechanistic link from breathing to neural coding, emotion, and cognition [48].

Conversely, respiratory instability is demonstrated in various disease states with a range of underlying contributory mechanisms including anatomic airway collapsibility, high loop gain, circulatory delay, load-capacity-drive imbalance, mechanical limitation, and central ventilatory control disruption. These unique parameters can be difficult to individually identify without specialized evaluation; thus, BSI may capture a shared endpoint of multiple pathophysiologic contributors, allowing for more nuanced characterization of abnormal sleep breathing beyond current discrete respiratory metrics.

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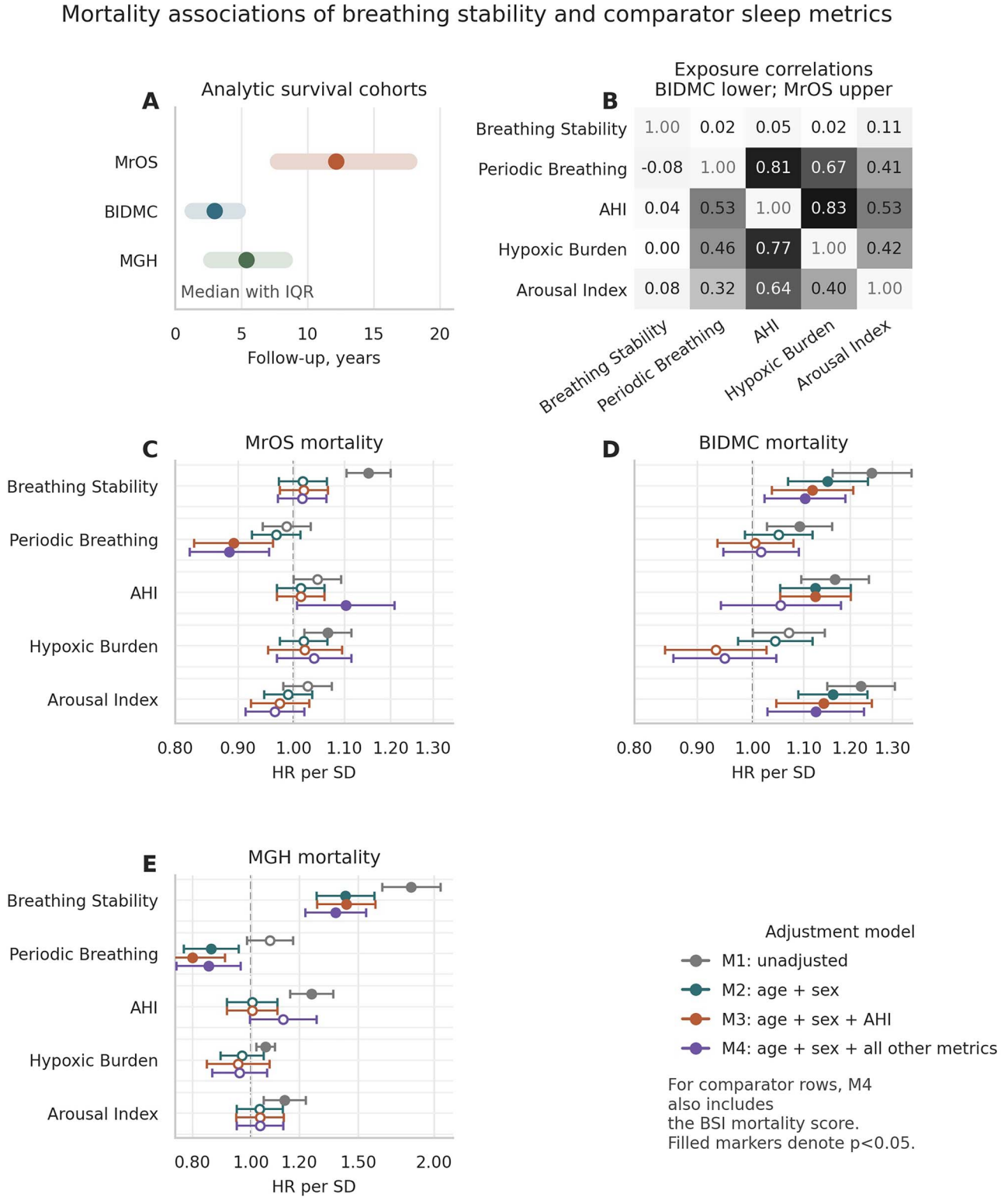


Figure 7. All-cause mortality associations of breathing stability and comparator sleep metrics. Panel A shows median follow-up and interquartile range for the analytic survival cohorts. Panel B shows Spearman correlations among the breathing-stability mortality score and comparator sleep metrics, with signed rho values and gray intensity reflecting absolute correlation magnitude; BIDMC is shown in the lower triangle and MrOS in the upper triangle. Panels C-E show cohort-specific Cox hazard ratios per 1-SD higher exposure for MrOS, BIDMC, and MGH. M1 is unadjusted, M2 adjusts for age and sex, M3 additionally adjusts for AHI, and M4 adjusts for age, sex, and all other sleep metrics; comparator M4 models also include the BSI mortality score. Filled markers denote $p < .05$; horizontal bars show 95% CIs.

The OSA/CSA/mixed phenotype analyses further refine the interpretation. BSI was higher in mixed and OSA-predominant phenotypes than in CSA-predominant phenotypes before full respiratory-burden adjustment, but these contrasts were attenuated or reversed after accounting for AHI and hypoxic burden. Median BSI increased with AHI severity within OSA, CSA,

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and mixed groups, and continuous models associated BSI to both obstructive and central event burden. Thus, BSI should not be interpreted as OSA-specific, nor as a direct loop-gain measurement. It is better viewed as an effort-derived instability marker that overlaps with, but remains distinct from, established approaches to ventilatory-control phenotyping and self-similar periodic breathing [12, 15, 18, 27, 38].

The cognition and disease analyses extend the physiologic validation into clinically relevant domains. Breathing-stability features showed cross-sectional prediction of cognitive scores, with stronger performance in MrOS than in MGH-Cog, and disease-label discrimination across most matched EHR-derived case–control sets. The age/sex matching makes the disease results particularly useful as a signal that BSI features are not merely recapitulating demographic risk. These findings suggest that instability contains clinically relevant information, but they do not establish directionality. Cognitive testing and disease labels were cross-sectional with respect to PSG timing, and breathing instability may be a consequence as well as a contributor to comorbid illness. Mechanistically, unstable breathing could reflect or promote sleep fragmentation, intermittent gas-exchange stress, autonomic fluctuation, or vascular stress, all of which are relevant to cognitive and cardiometabolic risk; however, this study did not directly measure blood gasses, cerebral hemodynamics, inflammation, or other intermediate pathways [40, 42].

The mortality findings support the premise that breathing stability should be treated as an overnight physiologic profile rather than a single summary value. The out-of-fold BSI feature score, which used the distribution and overnight organization of stable and unstable breathing, was associated with all-cause mortality in both clinical cohorts after adjustment for age, sex, AHI, hypoxic burden, arousal index, and periodic breathing/self-similarity, with the strongest adjusted association in MGH and a smaller but still significant association in BIDMC. The same score was attenuated after age adjustment in MrOS, indicating cohort dependence and likely differences in age structure, ascertainment, comorbidity, follow-up duration, and event distribution. Taken together, these results suggest that multifeature breathing-stability profiles may support survival-risk enrichment in some clinical settings, while requiring prospective validation before clinical decision use [2, 9, 10, 39–42, 49, 50].

Several processing considerations are important. The primary pipeline uses robust normalization of abdominal and thoracic Respiratory inductance plethysmography (RIP) belts before combining available effort channels. Resampling at 10 Hz is appropriate for the slow envelope modulations targeted here, but it is not designed to preserve every high-frequency signal feature. Paradoxical breathing is not separately annotated by the algorithm; it can influence BSI through the polarity-aligned belt combination and the morphology of the resulting effort envelope. Supplementary examples show that abdomen-only, thorax-only, and combined-belt BSI can differ during anti-phase breathing, underscoring that BSI is an effort-signal representation rather than a calibrated ventilation measure.

Other limitations temper interpretation. First, single-night PSGs and single-timepoint cognition or diagnosis labels introduce measurement variability, and EHR-derived disease labels were cross-sectional relative to PSG rather than uniformly incident endpoints. Second, residual confounding from medication use, treatment status, comorbid illness, and cohort-specific ascertainment may influence associations, including the cohort-dependent mortality-score results. Third, event-duration burden was not available in the merged analysis table and therefore could not be included as a mortality covariate. Fourth, posture data were available in the BDSP clinical cohorts but not across all cohorts; supine burden was associated with BSI, but posture adjustment did not materially change the posture-usable mortality sensitivity. Fifth, although sensitivity analyses supported similar BSI behavior across tested window lengths and overlap settings, we did not rerun the full final pipeline across every possible processing parameterization. Finally, no harmonized external RIP signal-quality score was available across cohorts; belt availability, robust normalization, clipping/outlier handling, and envelope safeguards reduce but do not eliminate signal-quality heterogeneity.

In summary, BSI provides a transparent, effort-derived measure of nocturnal breathing instability that complements AHI, hypoxic burden, desaturation frequency, arousal index, and periodic-breathing measures. Its value is that it makes the temporal organization of respiratory effort measurable across sleep stages and across the night, with demonstrated associations to physiology, cognition, EHR-derived disease labels, and cohort-dependent survival-risk scores. Future work should test night-to-night reliability, prospective outcome prediction, treatment responsiveness, and integration with EEG, cardiovascular, and wearable-derived sleep biomarkers.

Supplementary material

Supplementary material is available at *SLEEP* online.

Author contributions

WG served as the first author, overseeing data collection, algorithmic and software development, the design and execution of statistical analyses. Also prepared the initial full draft, generated the tables and figures, and coordinated the overall study workflow.

HS contributed to data acquisition, participated actively in group meetings, and provided critical feedback during the development process and manuscript revision.

RJT, and MBW, as co-senior authors, conceived the study, secured funding, and provided overarching leadership throughout the project. They oversaw all major aspects of data acquisition and analysis, offered strategic mentorship, and contributed in-depth critique during manuscript development.

Disclosure statement

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Non-financial disclosure: Dr. Westover is a co-founder, scientific advisor, and consultant to Beacon Biosignals and has a personal equity interest in the company. Dr. Thomas discloses (1) patent and license/royalties from MyCardio, LLC, for the ECG-spectrogram; (2) patent and license/royalties from DeVilbiss-Drive for an auto-CPAP algorithm; and (3) consulting for Jazz Pharmaceuticals, Guidepoint Global, and GLG Councils.

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Data availability

The MGH and BIDMC polysomnography data and associated mortality follow-up used in this study are available via the Brain Data Science Platform (BDSP) under the Human Sleep Project: https://bdsp.io/content/hsp/. MrOS polysomnography data are available through the National Sleep Research Resource (NSRR); access to additional MrOS outcomes data (including survival follow-up) is available via the MrOS Coordinating Center (San Francisco Coordinating Center) according to their data access policies. The analysis code for computing the BSI and reproducing the primary analyses is available at: https://github.com/bdsp-core/breathing-stability-index.

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